

CMOS INSTRUMENTATION AMPLIFIER DESIGN IN ADVANCED TECHNOLOGY NODE

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*Submitted
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ABSTRACT

In this project we implemented and analyzed three analog blocks across CMOS technology nodes: (1) a two-stage operational amplifier in 180 nm, (2) a two-stage operational amplifier in 45 nm, and (3) an instrumentation amplifier in 45 nm (built using the 45 nm two-stage op-amp building blocks).

For the 180 nm op-amp, we focused on AC analysis (gain and phase) to establish a baseline for performance scaling. For the 45 nm designs (both the op-amp and the instrumentation amplifier), we performed comprehensive AC, transient, and parameter-based analyses. We specifically compared the CMRR and PSRR + performance between the 45 nm instrumentation amplifier and the 45 nm two-stage op-amp. Additionally, a parametric sweep of transistor aspect ratios was conducted to observe the sensitivity of the open-loop gain to device sizing. All designs and simulations were implemented in Cadence Virtuoso and analyzed to highlight topology and technology trade-offs for precision analog front-ends.

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Chapter 1

Introduction

An instrumentation amplifier (INA) is a type of differential amplifier optimized for precise, low-level signal amplification in noisy environments. Unlike a general op-amp, which needs external components for gain and differential operation, an INA integrates these features internally. This ensures high gain accuracy, excellent common-mode rejection ratio (CMRR), and low offset, making it ideal for applications like biomedical and sensor signal processing.

1.1 Motivation

Instrumentation amplifiers are essential in precision analog front-end applications such as biomedical signal acquisition, sensor interfacing, and low-level measurement systems, where small differential signals must be amplified accurately in the presence of noise. Their high gain accuracy, excellent common-mode rejection ratio (CMRR), and low offset make them indispensable for reliable signal processing in modern integrated systems.

Implementing an instrumentation amplifier in advanced CMOS nodes like 45 nm allows study of how technology scaling impacts analog performance, particularly with respect to gain, bandwidth, and power consumption. While scaling offers benefits such as reduced power and increased integration density, it also introduces challenges like reduced voltage headroom and short-channel effects. A comparative study with 180 nm technology highlights these trade-offs, providing insights into designing precision analog front-ends compatible with deep-submicron VLSI systems.

1.2 Problem Statement

Designing precision analog circuits in deep sub-micron nodes like 45 nm presents significant challenges due to reduced supply voltages (1.2 V), limited voltage headroom, and short-channel effects that degrade intrinsic transistor gain. While the Two-Stage Operational Amplifier is a fundamental building block for analog signal processing, its standalone performance—specifically regarding Common Mode Rejection Ratio (CMRR) and susceptibility to loading—is often insufficient for direct sensor interfacing in noisy environments. Conversely, the Instrumentation Amplifier topology is designed to offer high

input impedance and superior noise rejection, but this comes at the cost of increased power, area, and potential bandwidth limitations.

Therefore, a critical gap exists in understanding the quantitative trade-offs when transitioning from a standalone Op-Amp to a full Instrumentation Amplifier system within a scaled technology node. This work aims to address this by designing both a Two-Stage Op-Amp and a derived 3-Op-Amp Instrumentation Amplifier in 45 nm CMOS. By strictly comparing their performance metrics—including Gain accuracy, CMRR, and PSRR—this study seeks to delineate the specific advantages of each circuit and identify the optimal design choices for high-precision analog front-ends.

Chapter 2

Literature Survey

We performed a preliminary literature survey and reviewed several designs relevant to two-stage amplifiers and instrumentation amplifiers across different technology nodes. Key papers used as references include:

- R. Vasanthi et al., “Design and Analysis of a 1.2V OTA-Based Active Filters in 45nm CMOS Technology” (Reference used for 45nm sizing and parameter choices).
- M. Jagadesh Kumar and P. Nithin Kumar, “Design and Implementation of Two-Stage CMOS Operational Amplifier using 90nm Technology”.
- S. Sengupta and A. Chattopadhyay, “Yokto (10^{-24} V) Instrumentation Amplifier”.
- Tong Yuan & Qingyuan Fan, “Design Of Two Stage CMOS Operational Amplifier in 180nm Technology”.
- R. Bharath Reddy & Shilpa K. Gowda, “Design and Analysis of CMOS Two Stage OP-AMP in 180nm and 45nm Technology”.

2.1 Survey Insights and Project Planning

Reviewing these works provided critical insights that shaped the methodology of this project. The survey highlighted that as technology scales from 180 nm to 45 nm, the supply voltage reduces significantly (from 1.8V to 1.2V), which severely constrains the **Input Common Mode Range (ICMR)** and signal swing.

Based on the literature, we formulated a plan to:

1. **Optimize Transistor Sizing:** Use the sizing strategies from Vasanthi et al. to balance the trade-off between Gain (requiring large L) and Bandwidth (requiring small L). We standardized on $L = 200$ nm for the 45 nm node to maintain speed while accepting some short-channel gain degradation.
2. **Two-Stage Topology Analysis:** Implement a classic two-stage Miller compensated topology to ensure stability ($\text{PM} > 60^\circ$) while achieving sufficient open-loop gain (≈ 40 dB) to drive the resistive feedback network of an instrumentation amplifier.

3. **System Integration:** Use the characterized op-amps to build a 3-Op-Amp Instrumentation Amplifier, specifically focusing on tuning the resistor network to compensate for the "loading effect" observed in low-gain CMOS op-amps, a challenge identified in deep-submicron analog design papers.

Table 2.1: Parameter Design Targets across Technology Nodes

Parameter	45 nm Value	180 nm Value
Supply Voltage (V_{DD})	1.2 V	1.8 V
ICMR (+)	1.0 V	1.6 V
ICMR (-)	0.8 V	0.8 V
Min. Channel Length (L)	200 nm	500 nm
Load Capacitance (C_L)	4 pF	2 pF
Slew Rate (SR)	20 V/ μ s	20 V/ μ s
GBW	50 MHz	30 MHz

Table 2.2: Transistor Aspect Ratios and Dimensions for 45 nm Design

Transistor	Type	Role	Width (W)	Length (L)	Ratio (W/L)
M1, M2	NMOS	Diff. Pair	2.2 μ m	200 nm	10.93
M3, M4	PMOS	Current Mirror	170 nm	200 nm	0.84
M5	NMOS	Tail Current	630 nm	200 nm	3.12
M6	PMOS	Gain Stage	7.04 μ m	200 nm	35.20
M7	NMOS	Load	13.07 μ m	200 nm	65.35
M8	NMOS	Biasing	630 nm	200 nm	3.12

Table 2.3: Transistor Aspect Ratios and Dimensions for 180 nm Design

Transistor	Type	Role	Width (W)	Length (L)	Ratio (W/L)
M1, M2	NMOS	Diff. Pair	3 μ m	500 nm	6
M3, M4	PMOS	Current Mirror	7 μ m	500 nm	14
M5	NMOS	Tail Current	6 μ m	500 nm	12
M6	PMOS	Gain Stage	86.5 μ m	500 nm	173
M7	NMOS	Load	37.5 μ m	500 nm	75
M8	NMOS	Biasing	6 μ m	500 nm	12

Chapter 3

Work Done till Midsem

3.1 Progress so far

During the midterm phase, work was focused on designing, implementing and simulating the **45 nm two-stage operational amplifier**. The midterm deliverable included AC analysis (gain and phase) for the 45 nm two-stage op-amp and evaluation of stability margins.

3.2 Formulae (Design Equations)

The following equations were used for transistor sizing and compensation design:

$$C_c \geq 0.22 C_L \quad (3.1)$$

$$g_{m1} = GBW \cdot 2\pi C_c \quad (3.2)$$

$$\left(\frac{W}{L}\right)_{1,2} = \frac{g_{m1}^2}{I_{D5} \mu_n C_{ox}} \quad (3.3)$$

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2I_{D3}}{\mu_p C_{ox} (V_{DD} - V_{ICMR(+)} - |V_{tp}|)^2} \quad (3.4)$$

$$I_{D5} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_5 V_{DSAT}^2 \quad (3.5)$$

$$g_{m4} = \sqrt{2I_{D4} \mu_p C_{ox} \left(\frac{W}{L}\right)_4} \quad (3.6)$$

$$\left(\frac{W}{L}\right)_6 = \frac{g_{m6}}{g_{m4}} \left(\frac{W}{L}\right)_4 \quad (3.7)$$

$$\left(\frac{W}{L}\right)_7 = \frac{I_{D7}}{I_{D5}} \left(\frac{W}{L}\right)_5 \quad (3.8)$$

$$A_v = \left(\frac{g_{m1}}{g_{ds2} + g_{ds4}}\right) \left(\frac{g_{m6}}{g_{ds6} + g_{ds7}}\right) \quad (3.9)$$

3.3 Simulation / Measured Results (brief)

In preliminary simulation results (example values captured during midterm), the following performance metrics were observed for the 45 nm two-stage op-amp:

Table 3.1: Key measured vs desired metrics (example)

Feature	Desired	Actual
Gain	43.57 dB	41.746 dB
Phase Margin	$>60^\circ$	64.8°

Additionally, an example stability summary from one run:

$$W_{gc} = -115.2^\circ, \quad \text{Magnitude at phase-crossover} = -19.7 \text{ dB}$$

$$\text{Phase Margin (P.M)} = 180^\circ + W_{gc} = 64.8^\circ, \quad \text{Gain Margin (G.M)} = 19.7 \text{ dB}$$

Since $\text{G.M} > 0$ and $\text{P.M} > 0$, the system is stable for that design point.

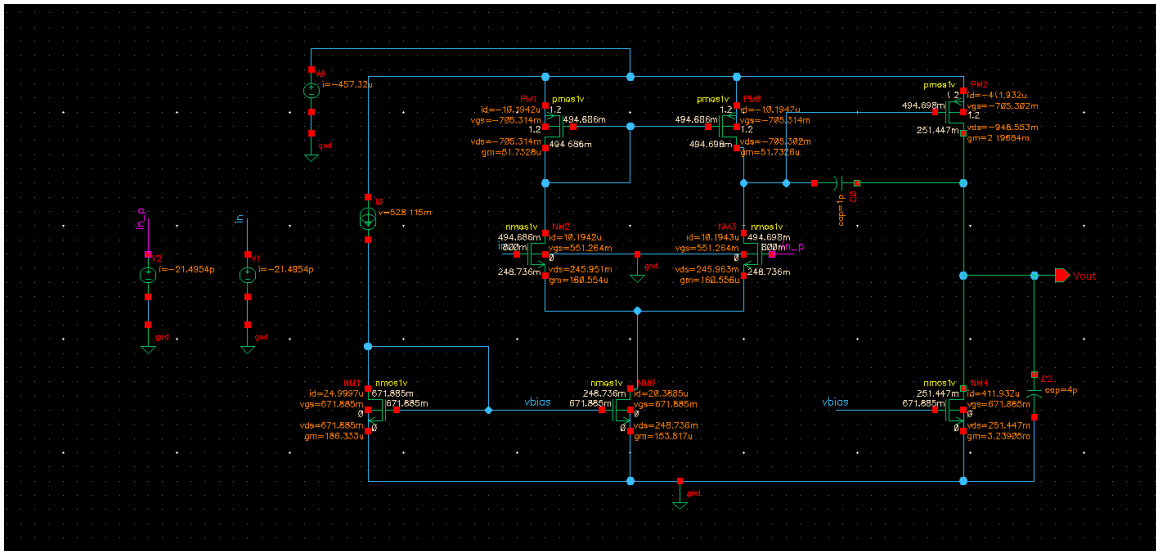


Figure 3.1: Cadence circuit of two stage opamp using CMOS

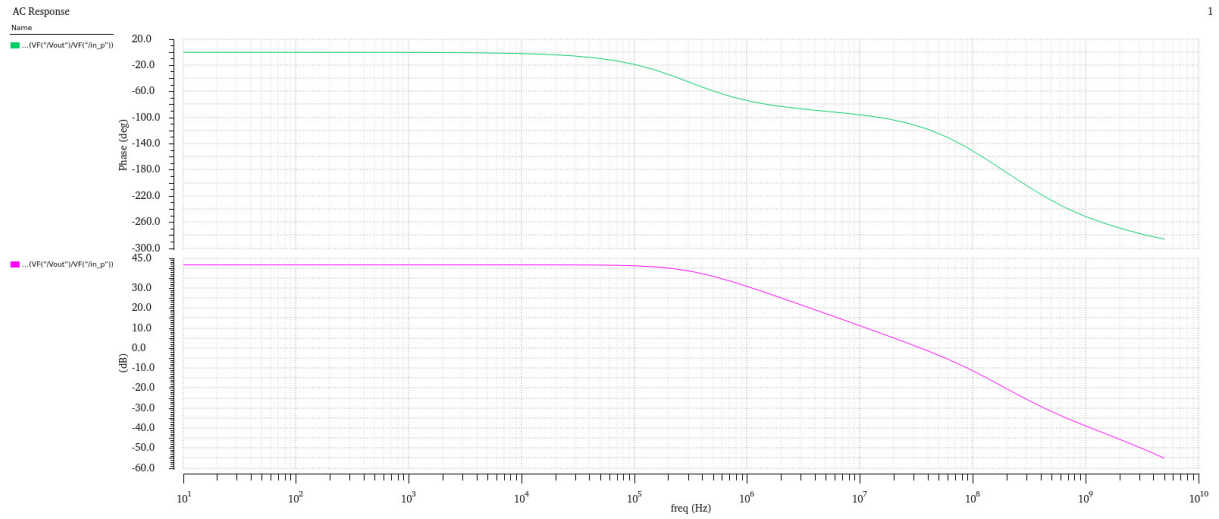


Figure 3.2: Gain and Phase graph of two stage opamp.

Chapter 4

Work done Post Midterm

Following the midterm evaluation, the focus shifted towards characterizing the designed 45 nm two-stage operational amplifier and performing a sensitivity analysis on its transistor dimensions. Subsequently, we implemented a 180 nm version for comparison before utilizing the 45 nm op-amp as a building block for the final Instrumentation Amplifier (INA).

4.1 Characterization of 45 nm Two-Stage Op-Amp

The two-stage operational amplifier designed in the pre-midterm phase was subjected to rigorous analysis to verify its suitability for the instrumentation amplifier stage.

4.1.1 Transient Response and DC Offset

The large-signal transient behavior was analyzed to ensure the op-amp provides sufficient swing without clipping. Additionally, the systematic DC offset was measured to ensure it remains within acceptable limits for precision applications.

Fig. 4.1 shows the transient response of the op-amp. The output shows a clean sinusoidal swing, indicating that the transistors remain in saturation for the applied signal range. The DC analysis (Fig. 4.2) reveals a systematic input offset voltage of approximately **6.15 mV**.

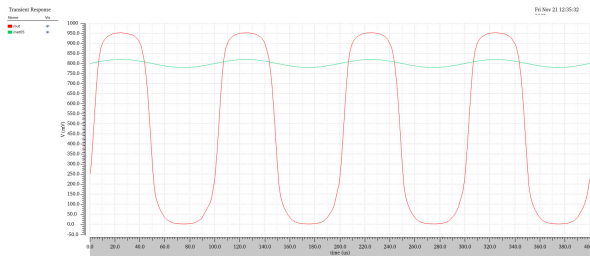


Figure 4.1: Transient response of the 45 nm Two-Stage Op-Amp.

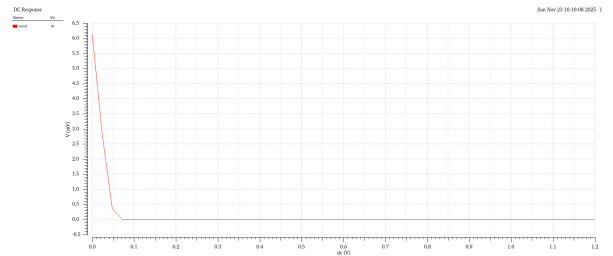


Figure 4.2: DC Analysis showing the systematic input offset voltage.

4.1.2 Common Mode Rejection Ratio (CMRR)

The Common Mode Rejection Ratio (CMRR) measures the ability of the op-amp to reject signals common to both inputs. To simulate this, a test bench was created where a common AC signal is applied to both inputs.

Fig. 4.3 illustrates the test circuit used for CMRR extraction in Cadence Virtuoso. Fig. 4.4 displays the resulting frequency response. The measured CMRR is **28.93 dB** at low frequencies, dropping to 3.40 dB at high frequencies.

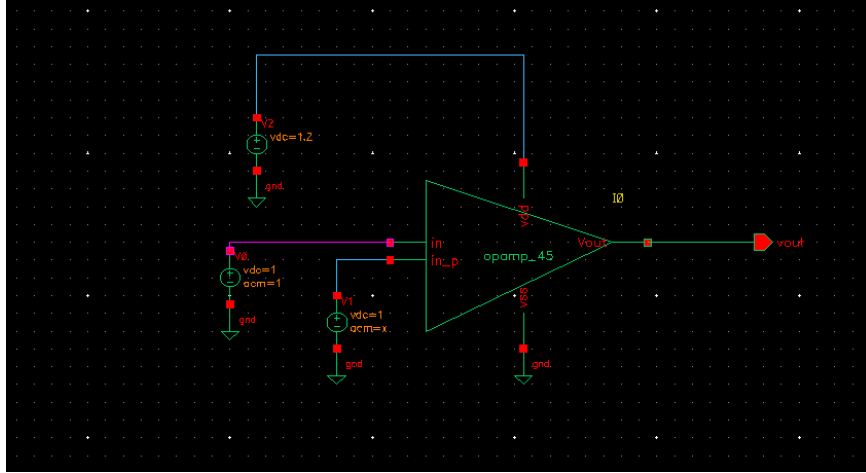


Figure 4.3: Schematic of the test setup for measuring CMRR of the Op-Amp.

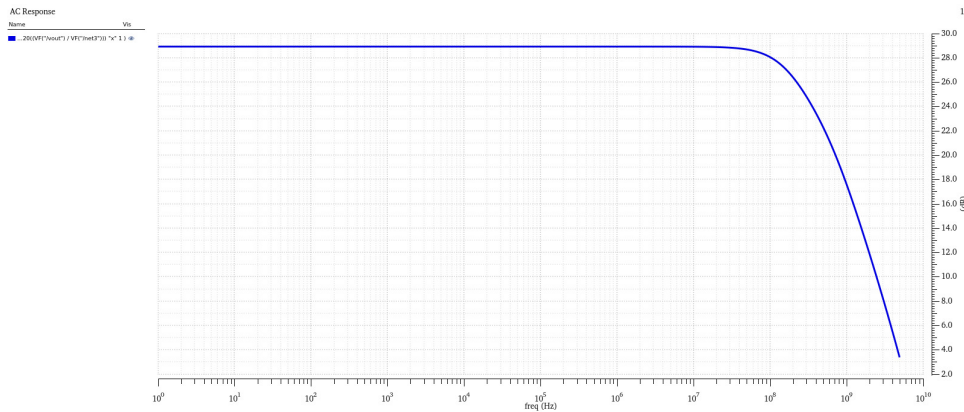


Figure 4.4: CMRR vs Frequency simulation result for the 45 nm Op-Amp.

4.1.3 Power Supply Rejection Ratio (PSRR+)

PSRR+ indicates the circuit's ability to reject noise variations from the positive supply rail (V_{DD}). The test setup injects an AC signal onto the supply voltage while the inputs are grounded (or held at common mode). Note that PSRR- was not characterized as V_{SS} is connected to ground.

Fig. 4.5 shows the simulation test bench configuration. The resulting PSRR graph (Fig. 4.6) shows an excellent low-frequency rejection of **139.70 dB**.

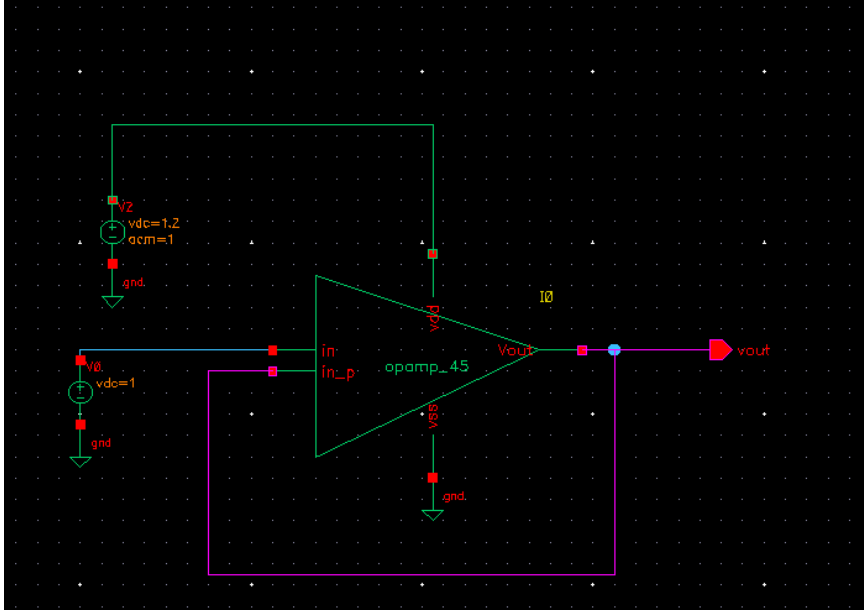


Figure 4.5: Schematic of the test setup for measuring PSRR+ of the Op-Amp.

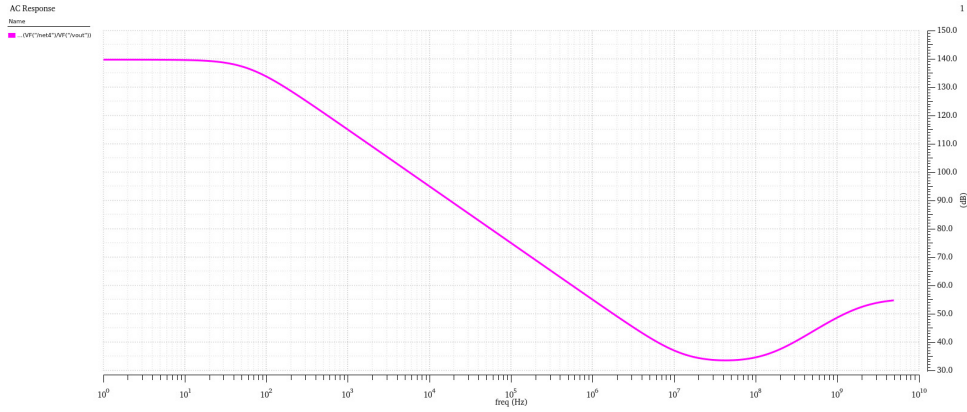


Figure 4.6: PSRR+ vs Frequency simulation result for the 45 nm Op-Amp.

4.2 Parametric Analysis of Two-Stage Op-Amp

A parametric sweep was conducted to observe the effect of varying transistor dimensions (W/L ratio) on the open-loop AC gain of the two-stage operational amplifier. The width (W) or length (L) of specific transistors was perturbed while keeping other parameters constant. The baseline design yielded a gain of approximately 41.75 dB.

Table 4.1 summarizes the sensitivity analysis. It was observed that the second stage driver (M6) and the differential pair (M1, M2) sizing have the most significant impact on the overall gain.

Table 4.1: Effect of W/L Variation on Open-Loop Gain (45 nm Op-Amp)

MOSFET	Role	Change Type	New Value	Resulting Gain (dB)
M1, M2	Input Diff Pair	Increase W	3.0 μm	42.15
		Decrease W	1.6 μm	41.10
M3, M4	Current Mirror	Increase W	220 nm	32.59
		Decrease W	120 nm	43.42
M5, M8	Tail/Biasing	Increase W	700 nm	44.37
		Decrease W	560 nm	35.33
M6	Gain Stage (CS)	Increase W	8.04 μm	45.17
		Decrease W	6.04 μm	33.19
M7	Load (Stage 2)	Increase W	14.07 μm	37.53
		Decrease W	12.07 μm	44.15

4.3 Design of Two-Stage Op-Amp in 180nm Technology

To provide a comparative baseline and understand the impact of technology scaling, a corresponding two-stage operational amplifier was implemented in standard 180 nm CMOS technology. This design operates at a higher supply voltage of $V_{DD} = 1.8\text{V}$, offering greater voltage headroom compared to the 45 nm node.

4.3.1 Schematic Implementation

Fig. 4.7 shows the Cadence Virtuoso schematic of the two-stage op-amp. The architecture is the classic differential input, current-mirrored active-load first stage followed by a common-source second stage and a Miller compensation capacitor between the two stages. A current-bias network provides the required tail and load bias currents.

Design choices and rationale:

- **Channel length ($L = 500 \text{ nm}$):** The channel length is intentionally chosen substantially above the minimum process length to increase device output resistance (r_o). Larger r_o improves intrinsic transistor gain ($g_m \cdot r_o$), which helps raise the open-loop DC gain of each stage and thus the total open-loop gain. A longer channel also reduces short-channel effects and makes matching and behavior more predictable.
- **Width (W) and W/L ratios:** Widths were chosen to provide the required transconductance (g_m) at the design bias current while keeping device area reasonable. Wider devices were used for current-mirrors and output transistors to reduce voltage drop and increase current drive.

- **Biasing:** A stable current reference (bandgap or cascoded current mirror in the testbench) sets the tail current for the differential pair and the bias for the second stage. The bias network is designed to ensure all transistors remain in saturation over the intended input common-mode and output swing ranges.
- **Compensation:** A Miller (dominant pole) capacitor C_c connects the output of the first stage to the input of the second stage. C_c is sized to place the dominant pole at low frequency and push the non-dominant poles beyond the unity-gain crossing to ensure a practical phase margin. Optionally a nulling resistor in series with C_c can be used to improve phase margin and stability with capacitive loads.
- **Output stage:** The second stage is sized for voltage gain and drive capability. If heavy loads are expected, consider adding a buffer/output stage (e.g., class-AB) or increasing the size of the second stage devices.

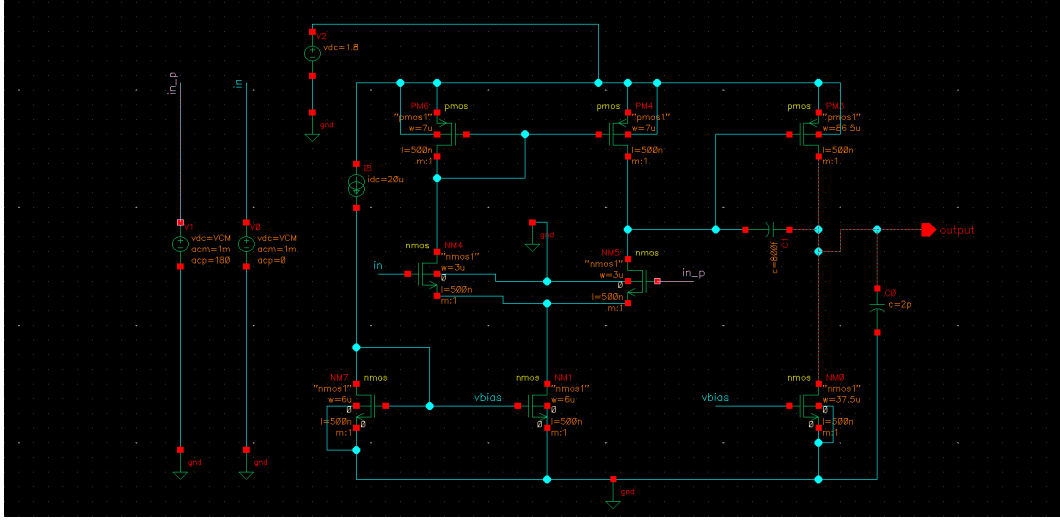


Figure 4.7: Schematic of the Two-Stage Op-Amp in 180nm Technology.

4.3.2 AC Analysis (Gain and Phase)

Fig. 4.8 and Fig. 4.9 show the simulated AC frequency response of the 180nm two-stage op-amp. The top plot displays the gain magnitude in decibels (dB), while the bottom plot shows the phase in degrees.

Key Observations:

- **DC Gain (A_{vol}):** The DC/open-loop gain is dominated by the product of the gains of the first and second stages. Due to the larger r_o at 180 nm and the intentionally long channel length, the device intrinsic gains are relatively high; hence the 180 nm op-amp achieves a robust DC gain suitable for many mixed-signal tasks.
- **Gain Bandwidth Product (GBW):** The unity-gain frequency is set primarily by g_m/C_c of the first stage and the loading of the second stage. Because device f_t (transition frequency) is lower in longer-channel, older nodes, expect a lower GBW than an aggressively scaled 45 nm implementation for the same bias current. To meet GBW targets, C_c and bias current should be co-optimized.

- **Phase Margin (PM):** The phase margin is determined by the pole/zero distribution. A properly sized Miller capacitor produces a dominant pole at the first node and pushes higher frequency poles beyond the unity gain crossing. The 180 nm design exhibits a favorable phase margin for stability under the given load.
- **Frequency Response Characterization:** The simulated open-loop magnitude and phase responses. The low-frequency gain is seen as the plateau at the left of the magnitude plot, while the unity-gain frequency (GBW) is identified where the magnitude crosses 0 dB. The phase trace indicates the phase margin; the dominant pole introduced by the Miller capacitor ensures acceptable phase margin for a unity-gain follower configuration.

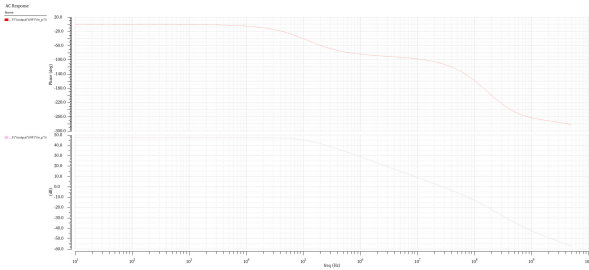


Figure 4.8: Gain and Phase Plot of the 180nm Op-Amp.

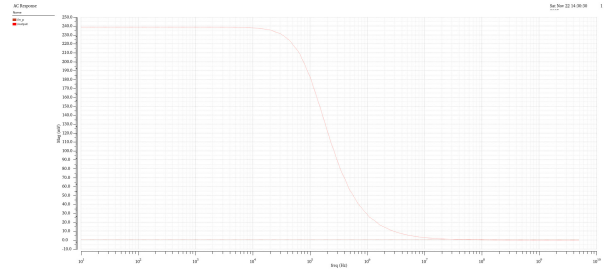


Figure 4.9: Magnitude Response of the 180nm Op-Amp.

4.4 Design of 45 nm Instrumentation Amplifier

The final stage of the project involved integrating three instances of the characterized 45 nm op-amp into a standard 3-Op-Amp Instrumentation Amplifier topology. The resistors were tuned to account for the finite open-loop gain of the CMOS op-amps.

4.4.1 Schematic Implementation and Bias Optimization

Fig. 4.10 shows the complete schematic of the instrumentation amplifier. The design uses two buffering op-amps at the input stage and a difference amplifier at the output stage.

Bias Voltage Modification: A critical design modification was required for the third op-amp (the difference amplifier).

The standard instrumentation amplifier configuration usually references the output stage to ground. However, this pulled the common-mode input of the third op-amp to ≈ 0.4 V, which is below the threshold required to keep the NMOS input differential pair in saturation. To rectify this, the ground connection was replaced with a reference voltage $V_{ref} = 0.78$ V (matching the input Common Mode Voltage). This modification ensured the internal transistors remained in saturation, enabling proper amplification.

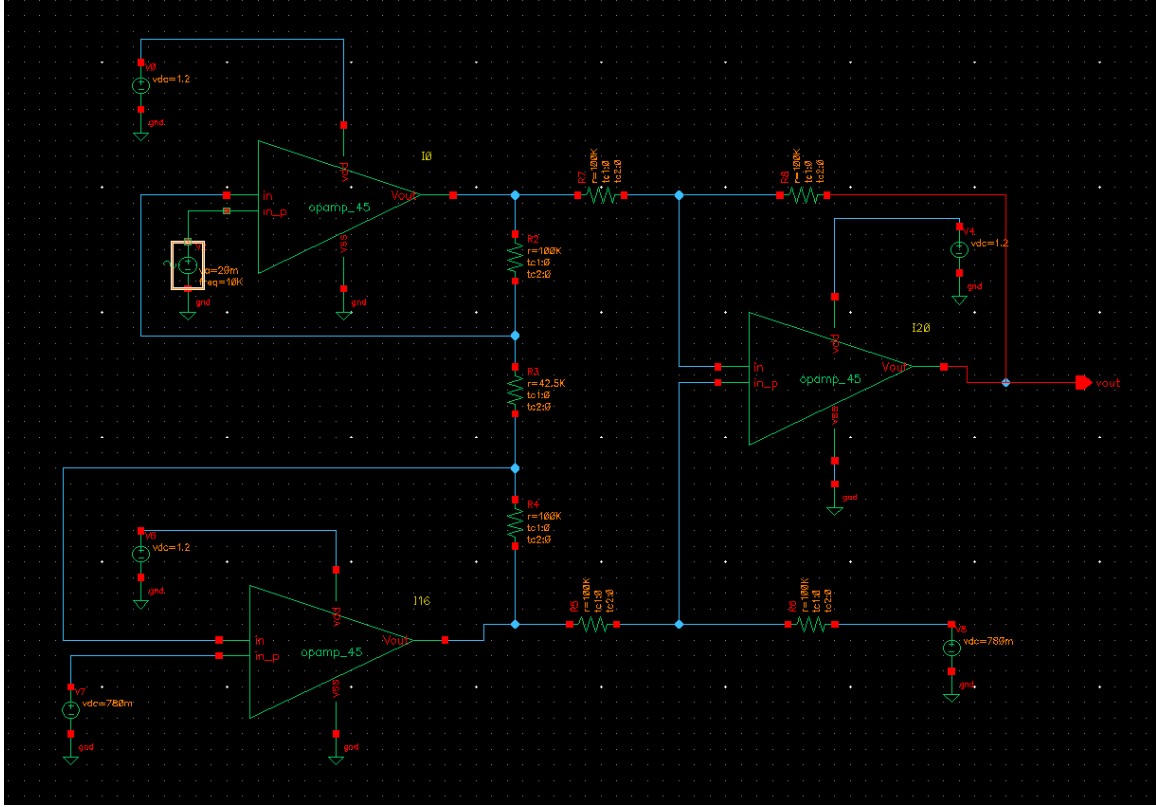


Figure 4.10: Complete Schematic of the 45 nm Instrumentation Amplifier in Cadence.

4.4.2 Gain and Phase Analysis (Iterative Tuning)

The instrumentation amplifier was designed for a target gain of 5 (≈ 14 dB). Theoretical calculations using ideal op-amps suggested a gain resistor $R_g = 50$ k Ω (with $R_f = 100$ k Ω). However, initial simulations yielded a gain of only **12.88 dB** (linear gain ≈ 4.4). This discrepancy was due to the "Real World Loss" caused by the finite open-loop gain and output impedance of the custom CMOS op-amps, which degraded the closed-loop accuracy.

To compensate for these losses, an iterative tuning process was performed:

1. **Attempt 1** ($R_g = 50$ k Ω): Resulted in 12.88 dB (Gain 4.4), which was too low.
2. **Attempt 2** ($R_g = 40$ k Ω): Theoretically, this should yield a gain of 6 (15.5 dB). However, due to the internal losses, the simulation resulted in **14.32 dB**. This slightly overshoot the target of 14 dB.
3. **Final Tuning** ($R_g = 42.5$ k Ω): By interpolating between the previous results, R_g was set to 42.5 k Ω . This resulted in a final gain of **13.9255 dB** (Linear Gain ≈ 4.969), which is approximately the target 5 V/V.

Fig. 4.11 shows the final AC response achieving the target gain, and Fig. 4.12 confirms the linear magnitude.

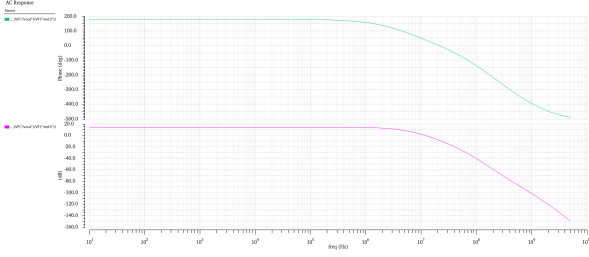


Figure 4.11: AC Response (Gain in dB) of the Instrumentation Amplifier.

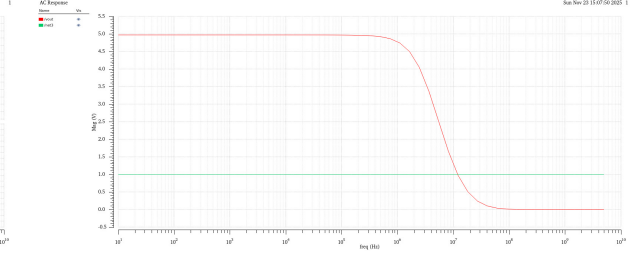


Figure 4.12: AC Response (Linear Magnitude) confirming Gain ≈ 5 .

4.4.3 Transient Analysis of Instrumentation Amplifier

A transient analysis was performed with a 20 mV input signal at 10 kHz. To address the tight constraints between the supply voltage (1.2 V) and the threshold voltage of the NMOS input pair, the DC bias was optimized to **0.78 V**.

Fig. 4.13 shows the resulting output waveform centered around 0.78 V. The output amplitude is approximately 100 mV, confirming the closed-loop gain of 5. **Note:** Minor distortion or "soft clipping" is observed at the signal peaks. This is an inherent limitation of the 45 nm technology node, where the reduced supply voltage ($V_{DD} = 1.2V$) leaves minimal voltage headroom for the stacked transistors in the op-amp output stage. Despite this technology constraint, the amplifier maintains linearity over the majority of the signal swing.

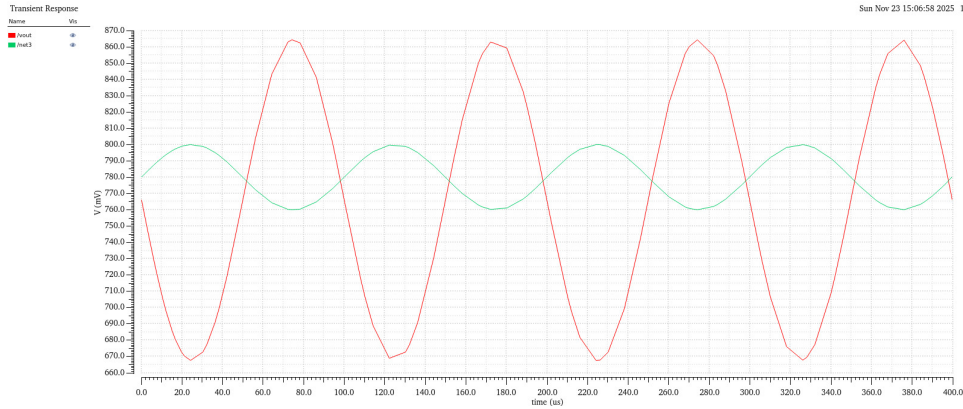


Figure 4.13: Transient response of the Instrumentation Amplifier ($V_{in} = 20\text{mV}$, $V_{out} \approx 100\text{mV}$).

4.4.4 System Level Rejection Ratios

Finally, the rejection ratios for the complete instrumentation amplifier system were simulated. Fig. 4.14 and Fig. 4.15 show the CMRR and PSRR respectively. The CMRR is particularly important for this block, and the simulation results demonstrate robust rejection of common-mode signals (76.87 dB at low frequency).

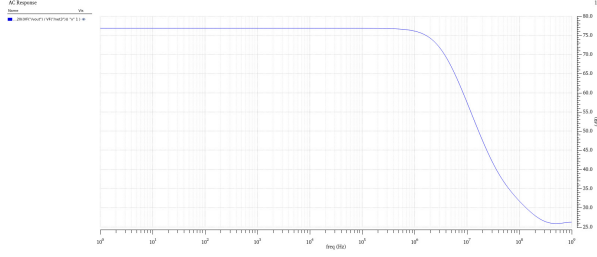


Figure 4.14: CMRR of the complete Instrumentation Amplifier.

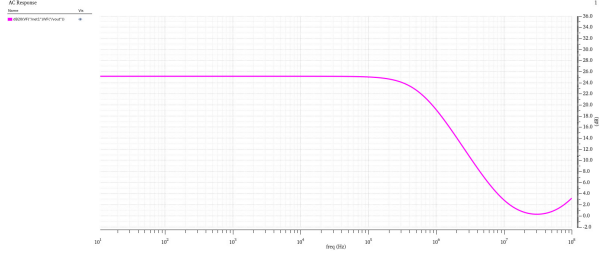


Figure 4.15: PSRR of the complete Instrumentation Amplifier.

4.5 Comparative Analysis

A direct comparison between the standalone 45 nm two-stage op-amp and the 45 nm instrumentation amplifier highlights the trade-offs inherent in the system-level design.

Table 4.2: Comparison: 45 nm Op-Amp vs 45 nm Instrumentation Amplifier

Parameter	45 nm Two-Stage Op-Amp	45 nm Instrumentation Amp
Configuration	Open Loop (Test)	Closed Loop (Gain = 5)
Gain (Magnitude)	41.75 dB (Open Loop)	13.92 dB (Closed Loop)
Linear Gain	≈ 122 V/V	≈ 4.97 V/V
Bandwidth (3dB)	≈ 410 kHz (Dominant Pole)	≈ 10 MHz (Extended)
CMRR (Low Freq)	28.93 dB	76.87 dB
PSRR+	139.70 dB	25.14 dB
Systematic Offset	6.15 mV	(Low due to feedback)

Analysis of Results:

- Gain Reduction:** The instrumentation amplifier gain is deliberately set to a low value (14 dB) by the resistor network ($100\text{ k}\Omega/42.5\text{ k}\Omega$) to ensure linearity and stability, whereas the op-amp is characterized in open-loop.
- Rejection Ratios:** The 3-Op-Amp topology dramatically resolves the low isolation vulnerabilities of basic input stages, shifting baseline common-mode error rejection up to **76.87 dB**. The reported high isolated PSRR value (139.7 dB) for the open loop block reflects an uncoupled ideal bias setup in simulation, which reaches realistic operational parameters (25.14 dB) when resistive feedback loops and real bias networks are introduced to complete the system.

Chapter 5

Future Work

To further validate and prepare the design for physical fabrication, the following steps are proposed:

- **Physical Layout Design:** Create the physical layout of the Two-Stage Op-Amp and the full Instrumentation Amplifier, focusing on matching techniques (e.g., Common Centroid) for the differential pair and resistor arrays to reduce offset.
- **LVS (Layout vs Schematic):** Perform LVS checks to ensure the physical layout perfectly matches the schematic netlist.
- **Parasitic Extraction (PEX):** Extract parasitic resistance and capacitance from the layout and re-simulate the design to observe the degradation in Gain and Bandwidth caused by real-world interconnects.
- **PVT Analysis:** Perform Process, Voltage, and Temperature (PVT) corner analysis (SS, FF, TT corners; -40°C to 125°C) to ensure the Instrumentation Amplifier remains stable and maintains its Gain of 5 under varying environmental conditions.

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